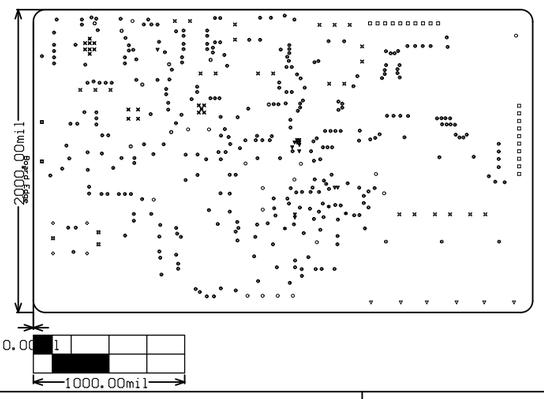
Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Overlay				
2	Top Solder	Solder Resist	0.40mil	3.5	
3	Top Layer	Copper	0.58mil		
4	Dielectric 1	FR-4 HIGH Tg	3.94mil	4.2	
5	GND1	Copper	1.18mil		
6	Dielectric2	FR-4 HIGH Tg	47.24mil	4.2	
7	PWR SPLIT PLANE	Copper	1.18mil		
8	Dielectric3	FR-4 HIGH Tg	3.94mil	4.2	
9	Bottom Layer	Copper	0.58mil		
10	Bottom Solder	Solder Resist	0.40mil	3.5	
11	Bottom Overlay				

NOTES:

1.THIS IS AN IMPEDANCE CONTROL BOARD.
2.TRACES WITH 4.9/11.1 MIL IN THE LAYER 01,TRACES WITH 5/11 MIL IN THE LAYER 4 ARE TO BE 100 OHMS +/- 10% IMPEDANCE CONTROLLED
3.TRACES WITH 6 MIL IN THE LAYER 01 , 4 AND 5 MIL TRACE WIDTH IN L02 ARE TO BE 50 OHMS +/- 10% IMPEDANCE CONTROLLED
4.TRACES WITH 4.25/11.75 MIL IN THE LAYER 02 ARE TO BE 100 OHMS +/- 10% IMPEDANCE CONTROLLED
5. ALL VIAS ARE TENTED. EXCEPT ON PAD VIAS AND 15 MIL VIAS ON TOP SIDE.
6.NON CONDUCTIVE VIA FILLING IS REQUIRED FOR ALL ON PAD VIAS
7. HARD GOLD PLATING REQUIRED FOR P1 CONNECTOR PADS ON TOP SIDE. FABRICATOR CAN DECIDE THE PLATING RANGE.



Symbol	Quantity	Finished Hole Size	Plated	Hole Type	Drill Layer Pair	Hole Length	Routed Path Length
▽	13	6.00mil (0.152mm)	PTH	Round	Top Layer - Bottom Layer	-	-
✕	17	7.87mil (0.200mm)	PTH	Round	Top Layer - Bottom Layer	-	-
⊙	316	10.00mil (0.254mm)	PTH	Round	Top Layer - Bottom Layer	-	-
✱	3	12.00mil (0.305mm)	PTH	Round	Top Layer - Bottom Layer	-	-
⬡	8	15.00mil (0.381mm)	PTH	Round	Top Layer - Bottom Layer	-	-
◘	2	19.69mil (0.500mm)	PTH	Slot	Top Layer - Bottom Layer	47.24mil (1.200mm)	27.56mil (0.700mm)
□	20	26.00mil (0.660mm)	PTH	Round	Top Layer - Bottom Layer	-	-
⊠	3	39.00mil (0.991mm)	NPTH	Round	Top Layer - Bottom Layer	-	-
○	17	40.00mil (1.016mm)	PTH	Round	Top Layer - Bottom Layer	-	-
⌘	24	40.16mil (1.020mm)	PTH	Round	Top Layer - Bottom Layer	-	-
▽	6	61.02mil (1.550mm)	PTH	Round	Top Layer - Bottom Layer	-	-
✲	3	78.74mil (2.000mm)	PTH	Round	Top Layer - Bottom Layer	-	-
◇	4	93.50mil (2.375mm)	NPTH	Round	Top Layer - Bottom Layer	-	-
	436 Total						

Slot definitions : Routed Path Length = Calculated from tool start centre position to tool end centre position.
Hole Length = Routed Path Length + Tool Size = Slot length as defined in the PCB layout

Drill tolerance:
For 6 mil blind via +0/-1 mil
For 6 mil via +0/-1 mil
For 7.87 mil via +0/-1 mil
For 10 mil via +0/-1 mil
For 12 mil via +0/-1 mil
For 15 mil via +0/-1 mil
For PTH +/- 3MIL
For NPTH +/-2 MIL

DESIGN INFORMATION

MIN. TRACK WIDTH: 4.25 MIL
MIN. CLEARANCE: 3.74 MIL
MIN. VIA PAD SIZE: 12 MIL
MINIMUM ANNULAR RING 0.05mm (2MIL) EXTERNAL
PER IPC-D-275 CLASS 2 LEVEL C
REGISTRATION TOLERANCES: METAL +/- 5 MIL, HOLES +/- 3 MIL
HOLE SIZE TOLERANCE (UNLESS OTHERWISE SPECIFIED): +/- 3 MIL

MATERIAL:
[ ] FR-408 [X] FR-4 High Tg [ ] OTHER
THICKNESS: [X] 62 MIL (1.6mm) +/-10% [ ] OTHER
TOLERANCE: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/-
BOW & TWIST: [X] ANSI IPC-6012 TYPE 3 CLASS 2
[ ] OTHER +/-
DRILLING:
REFERENCE: [X] AS SHOWN [X] NC_DRILL FILES
PTH COPPER THICKNESS: [X] 20-30 um [ ] OTHER
BOARD FINISH:
SILKSCREEN: [X] TOP [X] BOTTOM
SILKSCREEN COLOR: [X] WHITE [ ] OTHER
SOLDER RESIST COLOR: [X] GREEN [ ] OTHER
[X] MATTE [ ] SEMI-GLOSS
SURFACE FINISH: [X] IMMERSION GOLD (ENIG) [ ] ENEPIG
[ ] IMM. TIN/SILVER OR EQUIV [ ] OTHER
ARRAY/PANEL: [ ] CUT AND TRIM PER M1 BOARD OUTLINE
[ ] N.C. ROUTE [X] V. SCORE
CERTIFICATION: MATERIALS AND WORKMANSHIP FOR ALL PCBS TO MEET OR EXCEED THE REQUIREMENTS OF:
[X] ANSI IPC-A-600F CLASS -> [ ] 1 [X] 2 [ ] 3
[X] RoHS [ ] OTHER PER ORDER
ALL BOARDS MUST MEET OR EXCEED UL94-V0 REQUIREMENTS.
PCB MUST BEAR THE UL94V-0 UL REGISTERED MATERIAL ID NUMBER
ADDITIONAL REQUIREMENTS:
MICROSECTION: [ ] YES
BARE BOARD ELEC. TEST: [ ] NONE [X] REQUIRED [ ] PER ORDER

TEXAS INSTRUMENTS

PROJECT TITLE: AFE4950EVM
DESIGNED FOR: Public Release
FILE NAME: DC116_PCB.PcbDoc

ALL ARTWORK VIEWED FROM TOP SIDE	BOARD #: DC116E0	REV: A	SVN REV: Not In VersionControl
LAYER NAME = Drill Drawing1	TID #: N/A		
PLOT NAME = Fabrication Drawing	GENERATED : 9/25/2020 10:12:38 AM	TEXAS INSTRUMENTS	

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ENGINEER:	LAYOUT BY:
Midhun Raveendran	MANJUNATHA B
SCALE: 1.00	ALTIUM DESIGNER VERSION: 18.1.9.240